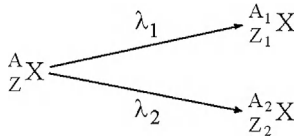
5.14. BRANCHING DECAY

Several times in this chapter the possibility of competing modes of decay has been noted; see e.g. Figure 1.2. In such competition, termed *branching decay* (see e.g. Fig. 5.5(g)), the parent nuclide may decay to two or more different daughter nuclides: e.g.



(5.71)

where for each branching decay a *partial decay constant* can be determined. These constants are related to the total observed decay constant for the parent nuclide as

$$\lambda_{\text{tot}} = \lambda_1 + \lambda_2 + \ldots \tag{5.72}$$

Each mode of decay in branching may be treated separately; the decay in an individual branch has a half-life based on the partial decay constant. Since only the total decay constant (the rate with which the mother nuclide, ${}^{A}_{Z}\text{X}$ in (5.71), decays) is observable directly, partial decay constants are obtained by multiplying the observed total decay constant by the fraction of parent decay corresponding to that branch. ${}^{64}\text{Cu}$ decays 43% by electron capture, 38% by negatron emission, and 19% by positron emission. The observed total decay constant is equal to $0.0541\ \text{h}^{-1}$ based on the half-life of 12.8 h. The partial constants are:

$$\lambda_{\text{EC}} = 0.43 \times 0.0541 = 0.0233\ \text{h}^{-1} \tag{5.73a}$$

$$\lambda_{\beta^-} = 0.38 \times 0.0541 = 0.0206\ \text{h}^{-1} \tag{5.73b}$$

$$\lambda_{\beta^+} = 0.19 \times 0.0541 = 0.0103\ \text{h}^{-1} \tag{5.73c}$$

These partial decay constants correspond to partial half-lives of 29.7 h for electron capture decay, 33.6 h for β^- decay, and 67.5 h for positron decay.

5.15. SUCCESSIVE RADIOACTIVE DECAY

There are many instances where a parent decays to a daughter which itself decays to a third species (i.e. a "grand-daughter"). The chains of radioactive decay in the naturally occurring heavy elements include as many as 10−12 successive steps (Fig. 1.2).

$$\text{X}_1 \xrightarrow{\lambda_1} \text{X}_2 \xrightarrow{\lambda_2} \text{X}_3 \xrightarrow{\lambda_3} \text{X}_4 \ldots \tag{5.74}$$